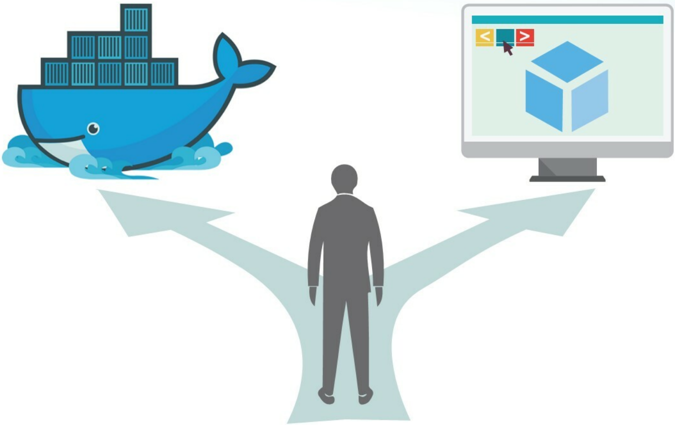
How Docker Differs *from a* Virtual Machine

Often, people get confused between Docker and a virtual machine. This article explains how Docker differs from a virtual machine and why you should choose the former over the latter.



The virtualisation concept originated in the 1970s during the mainframe era when IBM was putting in a lot of effort to create efficient time-sharing solutions. This implied sharing computer resources among multiple users and trying to achieve high performance and high utilisation. Today's data centres use virtualisation technologies to create an abstraction of physical hardware. Physical resources are divided into many logical resources and different users leverage the benefits of this. Here, the resources include CPUs, memory, storage, networks and applications.

What is virtualisation?

Virtualisation is creating a virtual version of something (like a server, storage, OS or networking resources), which is similar to the actual version.

What is a virtual machine?

In layman terms, a virtual machine is the logical replica of a physical machine. We can say that it's an emulation of the physical system. It follows the same architecture and functionality that the physical system provides.

Different types of virtualisation

There are many types of virtualisation, but we look at just the major types here.

Application virtualisation: Users can access different applications from remotely located servers instead of their